

Prosecuting Inequity: Climate Investment, Social Decay, and the Architecture of Equitable Decarbonization

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Abstract

This analysis integrates institutional economics, policy architecture, and techno-economic modeling to examine how climate investment structures systematically produce inequitable outcomes while undermining decarbonization effectiveness. We document three converging failures: (1) clean energy subsidies concentrating among highest-income quintiles despite nominal equity objectives, (2) categorical technology framing that obscures lifecycle emission dependencies and enables incumbent rent-seeking, and (3) erosion of foundational institutional capacity—educational systems, digital infrastructure, economic opportunity—that prevents disadvantaged populations from participating in energy transitions regardless of program access.

The analysis proceeds in three integrated parts. Part I establishes the institutional and fiscal context, documenting dollar reserve currency decline, middle-class economic compression, educational capacity collapse, and strategic resource misallocation exemplified by programs like Golden Dome. These constraints frame feasible policy space. Part II presents comparative techno-economic analysis of biomass-to-hydrogen pathways, demonstrating that modular gasification clusters can achieve competitive levelized costs (\$9.57–\$10.87/kg at 100-unit scale) while enabling distributed rural participation. Part III synthesizes policy architecture guardrails: parliament-based allocation resistant to incumbent capture, reusable capital structures enabling multiple project cycles from single appropriations, performance-based accountability requiring demonstration rather than assumption of benefits, and mandatory lifecycle transparency eliminating categorical framing advantages.

The electric vehicle case study reveals the mechanism. "Zero emission" classification allows vehicles manufactured with fossil-intensive processes and charged on coal-dominated grids to receive identical subsidies and environmental credit as vehicles with genuinely low lifecycle emissions. This categorical framing benefits incumbent utilities (increased electricity demand regardless of generation mix), natural gas producers (marginal generation), and established manufacturers, while suppressing recognition that 11 tonnes CO₂e per vehicle in manufacturing emissions—reducible by 92% through available renewable-powered battery production, green steel, and responsible mining—remains systematically unaddressed because all three actor classes (commercial, policy, civic) satisfy their objectives through symbolic classification rather than measured performance.

Biomass-derived hydrogen offers a contrasting pathway: lifecycle emissions depend directly on feedstock sustainability and process energy rather than external grid composition, preventing the opacity that enables three-actor evasion. Infrastructure requirements create barriers but also opportunities for distributed ownership models serving rural feedstock sources. The policy challenge lies not in technical feasibility—demonstrated through detailed process modeling—but in allocation mechanisms that counteract incumbent advantages and benefit concentration.

The analysis concludes that equitable decarbonization requires institutional innovations targeting benefit distribution rather than aggregate investment levels: mandatory lifecycle labeling eliminating categorical framing, subsidy scaling tied to demonstrated emission reductions rather than technology classification, infrastructure investment parity across pathways to enable comparative assessment, and deliberative allocation processes incorporating stakeholder representatives rather than technocratic optimization. Without these structural countermeasures, climate investment will replicate rather than remediate the systematic inequities documented across broadband deployment, educational infrastructure, and existing clean energy programs.

Keywords: Climate policy architecture; Lifecycle carbon accounting; Biomass gasification; Electric vehicle emissions; Incumbent rent-seeking; Reusable capital structures; Distributed energy systems; Institutional capacity; Categorical framing; Technology assessment